S M MAHAMUDUL HASAN

APPLIED AI ENGINEER | LLM APPLICATIONS, RAG & NLP | PYTHON, FASTAPI, PYTORCH

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PROFESSIONAL SUMMARY

Applied AI engineer with 3+ years of production software experience, an MSc in AI and Robotics (2026) and three peer-reviewed IEEE and Springer papers. Builds LLM applications end to end: a self-hosted bilingual chat service on Llama 3.2 with 9 FastAPI endpoints, NDJSON streaming, session memory, health checks and pytest suites. Brings deep NLP experience, from an 11,434-article labelled dataset to fine-tuned Bangla BERT at 0.96 F1, plus enterprise API and SAP data integration.

TECHNICAL SKILLS

LLM and RAG: LLM applications, RAG, prompt and context engineering, Llama 3.2 (Ollama), session memory, streaming
NLP: BERT fine-tuning, text classification, sentiment analysis, information extraction, Word2Vec, GloVe, LSTM
ML and Evaluation: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CNN, feature engineering, baselines, F1
Back End and Data: Python, FastAPI, Flask, REST APIs, validated schemas, SQL, Pandas, NumPy, R, Java, TypeScript
MLOps and Tools: Docker Compose, Git, GitHub Actions, GitLab CI/CD, Azure DevOps, pytest, Jira, Claude Code, n8n
Enterprise and Robotics: SAP data integration, Pricefx (certified), Groovy, SAP CPQ, Unqork, Unity/C#, NetLogo

PROFESSIONAL EXPERIENCE

Applied AI Engineer | BritAI Labs | Jan 2026 - Present

London, UK | Fixed-term contract

- Delivered a private bilingual LLM chat service with 9 REST endpoints and real-time NDJSON token streaming by building FastAPI routes on Llama 3.2 via Ollama with validated request/response schemas and Docker Compose.
- Kept multi-turn chats coherent with a 20-message context window per session by building session memory, Bangla/English system prompts and JSON/Markdown export.
- Made every release testable without a running model by writing pytest unit and API tests against a fake LLM client and adding liveness and model-readiness health checks.

Application Engineer | All Generation Tech | Jun 2024 - Sep 2024

Remote, United States | Contract

- Improved system efficiency by 20% by integrating APIs and JSON data flows between client systems.
- Cut deployment time by 30% by translating complex user stories into backend applications on the Unqork platform.

PRICEFX ENGINEERING EXPERIENCE

Pricefx Configuration Engineer | Eglobal Infotech | Nov 2023 - Apr 2024 | Contract | Remote, United States | Client: Canidium
Pricing Software Engineer | Ark Solutions | Jun 2022 - Sep 2023 | Full-time | Remote, Australia

- Reduced manual pricing effort by 95% by automating SAP data migration and data-source integrations in Pricefx.
- Cut pricing-processing time by 25% and operational errors by 15% by optimising Groovy rules and leading unit testing.
- Delivered three B2B Pricefx implementations in three countries by translating requirements into CPQ workflows.

Machine Learning and NLP Research Apprentice | HAB Lab | Feb 2020 - May 2022

HCI AI and Blockchain Laboratory | Sylhet, Bangladesh | On-site

- Reached 0.96 F1 on Bangla fake-news detection by labelling 11,434 news articles, benchmarking five models up to fine-tuned Bangla BERT, and serving predictions through a Flask API to an app adopted by a media agency.
- Built and evaluated ML, deep-learning and NLP models for classification, sentiment analysis and information extraction, using Word2Vec and GloVe embeddings, LSTM and BERT.
- Co-authored three IEEE and Springer papers by leading a five-member team and benchmarking models against baselines.

WordPress Web Developer | InnovexIT | Sep 2018 - Feb 2020

Remote | Part-time

- Raised client organic traffic by 40% by launching 15+ e-commerce and corporate websites and improving site speed and SEO.

SELECTED PROJECTS

Local LLM Chat API | Private Bangla/English chat on Ollama: FastAPI, NDJSON streaming, 20-message memory, export, Docker Compose, pytest
RoadReady | Live bilingual theory-test web app: 648 questions, 165 rules, timed mock tests; TypeScript, Vite, GitHub Actions
Autonomous Robot Simulation | Unity/C# robot using raycast sensors, dynamic steering and adaptive motor force to handle obstacles and slopes

SELECTED PUBLICATIONS

A Benchmark of Machine Learning and Deep Learning Algorithms for Detecting Fake News in Bangla Language | Second author | IEEE Xplore | **Code**
Determining the Most Effective Machine Learning Techniques for Detecting Phishing Websites | First author | Springer Nature | **Code**
Can Machine Learning Techniques Predict Prostate Cancer Accurately? The Fact and Remedy | First author | IEEE Xplore | **Code**

EDUCATION

MSc in Artificial Intelligence and Robotics | University of Hertfordshire | Hatfield, UK | Sep 2024 - Sep 2026
BSc in Computer Science and Engineering | Shahjalal University of Science and Technology | Sylhet, Bangladesh | Jan 2017 - Dec 2022
Awards | Top 5 of 400+ teams, National Hackathon on Frontier Technology, Govt of Bangladesh (2021) | Top 10, SigmaHacks (2022)